

Inferring effective interactions in a multispecies oral biofilm model: Bayesian calibration and a testable prediction

Keisuke Nishioka^{1,2,3,4*}, Felix Klempt¹, Hendrik Geisler¹,
Rumjhum Mukherjee^{2,3}, Nils Heine^{2,3}, Katharina Doll-Nikutta^{2,3},
Meike Stiesch^{2,3}, Szymon P. Szafranski^{2,3}, Meisam Soleimani¹,
Philipp Junker¹

¹*Institute of Continuum Mechanics, Leibniz Universität Hannover, An
der Universität 1, Garbsen, 30823, Germany.

²Department of Prosthetic Dentistry and Biomedical Materials Science,
Hannover Medical School, Hannover, Germany.

³Lower Saxony Centre for Biomedical Engineering, Implant Research and
Development (NIFE), Hannover, Germany.

⁴School of Science for Open and Environmental Systems, Graduate
School of Science and Technology, Keio University, Yokohama, Japan.

*Corresponding author(s). E-mail(s):

keisuke.nishioka@stud.uni-hannover.de;

Abstract

We present a GPU-accelerated Bayesian framework for inferring the interaction parameters of a five-species oral biofilm model derived from the extended Hamilton principle. The symmetric 5×5 interaction matrix is parametrised by its 15 independent entries. Inference proceeds in stages: the likelihood first contains the species composition only, and the community viability is then added as a further channel. The forward model is compiled with JAX and evaluated for all particles in parallel on a GPU, giving a $\sim 150\times$ speed-up over the CPU baseline. No interaction parameter is fixed to zero a priori, and every inference stage is checked with independent seeds against explicit convergence criteria. Applied to in-vitro time-course data of Heine et al. under four combinations of community state (commensal vs. dysbiotic) and cultivation (static vs. dynamic, the latter in a HOBIC flow reactor), the inferred effective parameters reproduce the observed trajectories, including the late *P. gingivalis* (Pg) increase

under dysbiotic dynamic cultivation (RMSE [0.05–0.12]). The *F. nucleatum* (Fn)–Pg coefficient is positive in that condition (90% interval [[+0.8, +5.2]]), positive under dysbiotic static cultivation only within the dominant of two posterior modes, and indistinguishable from zero in both commensal conditions. The inferred model reproduces the measured pH under dynamic cultivation, which is not used in the inference ($R^2 = [0.78]$ through a fixed regression on the species fractions). Posterior predictive simulations without Fn suppress the late Pg increase with probability [0.87–0.90], yielding a directly testable prediction.

Keywords: Bayesian inference, transitional Markov chain Monte Carlo, oral biofilm, multi-species interaction, parameter identification, peri-implantitis

1 Introduction

Multi-species biofilms that form on dental implant surfaces are a principal etiological factor in peri-implantitis (Kolenbrander et al. 2010; Marsh 2003). Recent *in vivo* characterisations indicate that these microbial communities are highly complex, individualized, and dynamically structured over time (Dieckow et al. 2024; Cui et al. 2025). In a controlled *in vitro* study, Heine et al. (2025) investigated the temporal dynamics of five representative oral taxa, *Streptococcus oralis* (So), *Actinomyces naeslundii* (An), *Veillonella* spp. (Vei; *V. dispar* in commensal, *V. parvula* in dysbiotic conditions), *Fusobacterium nucleatum* (Fn), and *Porphyromonas gingivalis* (Pg), across four experimental conditions defined by community state (commensal vs. dysbiotic) and cultivation mode (static vs. HOBIC reactor). Importantly, the two community states incorporated distinct *Porphyromonas gingivalis* strains: the commensal consortium included the avirulent strain Pg DSM 20709, whereas the dysbiotic consortium employed the virulent strain Pg W83, which harbors a full complement of gingipain and fimbrial virulence determinants (Heine et al. 2025). Building on the extended Hamilton principle (Junker and Balzani 2021), Klempt et al. (2026) formulated a continuum model in which a symmetric interaction matrix $\mathbf{A}$ governs species growth, with an optional decay vector $\mathbf{b}$ for antibiotic effects. Fritsch et al. (2026) subsequently introduced Bayesian updating with surrogate models for biofilm parameter estimation under hybrid uncertainties.

Despite these advances, estimating the 15-dimensional parameter vector (15 interaction parameters) from limited time-course data remains challenging. Two principal difficulties arise: (i) poor identifiability when data are sparse relative to the parameter dimension—with $N_t = 5$ usable time points (Day 1 is consumed as initial condition) and 5 species, one has at most 25 observations for 15 unknowns—and (ii) multimodal posterior landscapes caused by the coupled ϕ_i – ψ_i dynamics, which render standard MCMC methods inefficient (Robert and Casella 2004).

In this work we address these challenges through three complementary strategies. First, we introduce the data in two phases: the interaction matrix is first inferred from the species composition alone, and the viability measurements are added only once the composition-based posterior is available. Second, we compile the forward model

in JAX (Bradbury et al. 2018) and evaluate the entire particle ensemble on GPU via
 batched `vmap`, achieving a $\sim 150\times$ speedup over the CPU baseline. Third, we infer
 the posterior with TMCMC (Ching and Chen 2007; Betz et al. 2016) in stages with
 progressively narrowed prior boxes and verify every stage with independent seeds
 and explicit convergence criteria (Section 4.4). The framework is applied to all four
 experimental conditions of Heine et al.; it reproduces commensal homeostasis as well
 as the late pathogen increase of dysbiotic biofilms under HOBIC flow, and it yields a
 prediction that can be tested by a leave-one-species-out experiment.

2 Mathematical model

2.1 State variables and kinematics

Following Klempt et al. (2026, 2024), the biofilm is described at a material point by
 two sets of internal variables: the volume fraction $\phi_i \in [0, 1]$ occupied by species i
 ($i = 1, \dots, n$; here $n = 5$), and the viability fraction $\psi_i \in [0, 1]$ representing the
 proportion of living cells within species i . The living volume fraction is then

$$\bar{\phi}_i := \phi_i \psi_i, \quad i = 1, \dots, n. \quad (1)$$

An additional void fraction ϕ_0 closes the system via the holonomic constraint

$$\sum_{l=0}^n \phi_l(t) - 1 = 0 \quad \forall t. \quad (2)$$

2.2 Variational formulation

The evolution equations are derived from the principle of the minimum of the dis-
 sipation potential (Junker et al. 2014; Hackl 1997), the local form of the extended
 Hamilton principle (Junker and Balzani 2021; Junker et al. 2025):

$$\dot{\Psi} + \Delta + C \rightarrow \text{stat}_{(\xi, \gamma)}, \quad (3)$$

where $\xi = (\phi, \psi)^T$ collects the volume fractions and viability fractions, γ is a Lagrange
 multiplier enforcing the volume constraint, C is the constraint function, and Δ the
 dissipation function. The free energy density encodes nutrient-driven growth and
 antibiotic-induced killing:

$$\Psi = -\frac{1}{2} c^* \bar{\phi}^T \mathbf{A} \bar{\phi} + \frac{1}{2} \alpha^* \psi^T \mathbf{B} \psi, \quad (4)$$

where c^* is the nutrient concentration, α^* the antibiotic concentration, $\mathbf{A} \in \mathbb{R}^{n \times n}$
 the symmetric growth coefficient matrix, and $\mathbf{B} = \text{diag}(b_1, \dots, b_n)$ the antibiotic
 sensitivity matrix.

The dissipation function is derived from the rate-dependent potential (Klempt
 et al. 2026):

$$\Delta = \frac{1}{2} \dot{\phi}^T \boldsymbol{\eta} \dot{\phi} + \frac{1}{2} \dot{\psi}^T \boldsymbol{\eta} \dot{\psi}, \quad (5)$$

with the diagonal viscosity matrix $\boldsymbol{\eta} = \text{diag}(\eta_1, \dots, \eta_n)$. The first term couples $\dot{\phi}_i$ and $\dot{\psi}_i$ through $\dot{\bar{\phi}}_i = \dot{\phi}_i \psi_i + \phi_i \dot{\psi}_i$, giving rise to non-linear dissipative interactions; the second term regularises the system to full rank. Following [Klempt et al. \(2026\)](#), the viscosity parameters are set to $\eta_i = 1$ for all species, rendering them dimensionless scaling factors; the physical time scale is absorbed into the time-step size Δt .

The volume constraint (2) enters through a Lagrange multiplier γ :

$$C = \gamma \left(\sum_{l=0}^n \phi_l - 1 \right). \quad (6)$$

2.3 Stationarity conditions

Evaluating the stationarity conditions yields the coupled evolution equations ([Klempt et al. 2026](#)):

Volume fraction ($i = 1, \dots, n$):

$$0 = -c^* \psi_i (\mathbf{A} \bar{\boldsymbol{\phi}})_i + \eta_i (\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i) + \gamma, \quad (7)$$

Viability fraction ($i = 1, \dots, n$):

$$0 = -c^* \phi_i (\mathbf{A} \bar{\boldsymbol{\phi}})_i + \alpha^* b_i \psi_i + \eta_i (\dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i) + \gamma, \quad (8)$$

Void fraction:

$$0 = \gamma + \dot{\phi}_0, \quad (9)$$

Volume constraint:

$$0 = \sum_{l=0}^n \phi_l - 1, \quad (10)$$

In the numerical implementation, a logarithmic barrier penalty Ψ_{bar} with $K_p = \eta_i \times 10^{-4}$ is added to enforce the box constraints $\phi_i, \psi_i \in (0, 1)$ ([Klempt et al. 2026](#)); its contribution is omitted from Eqs. (7)–(9) for clarity. The system (7)–(10) constitutes $2n + 2$ coupled non-linear equations for the unknowns $(\phi_1, \dots, \phi_n, \phi_0, \psi_1, \dots, \psi_n, \gamma)$, solved at each time step via Newton’s method with an implicit Euler discretisation ($\Delta t = 10^{-4}$, $N_{\text{step}} = 2,500$ steps, $t_{\text{max}} = 0.25$ corresponding to the 21-day experimental window).

The variational derivation automatically ensures symmetry $A_{ij} = A_{ji}$ and thermodynamic consistency ([Junker and Balzani 2021](#)). Since c^* enters the evolution equations only through the product $c^* A_{ij}$, its absolute value is non-identifiable; we fix $c^* = 25$ so that the inferred A_{ij} remain $\mathcal{O}(1)$. The Heine et al. experiments do not involve antibiotic agents, so we set $\alpha^* = 0$, which eliminates the decay term $\alpha^* b_i \psi_i$ from Eq. (8); the antibiotic sensitivity matrix $\mathbf{B}$ is therefore inactive and excluded from estimation. The symmetric interaction matrix $\mathbf{A}$ has 15 independent entries (Section 2.4); these form the 15-dimensional parameter vector $\boldsymbol{\theta}$. Removing a species from the consortium is represented by switching it off (its volume fraction is held at zero), not by a zero initial value: in this formulation a species with a vanishing initial fraction can regrow from numerical floor values.

2.4 Symmetric interaction matrix

The symmetry $A_{ij} = A_{ji}$ arises automatically from the quadratic form of the free energy (4), reducing the off-diagonal degrees of freedom from 20 to 10. Together with the 5 diagonal entries, the symmetric matrix has 15 independent entries. These are distinct from the antibiotic sensitivity parameters b_i in Eq. (8), which vanish when $\alpha^* = 0$. The resulting 15-dimensional parameter vector is

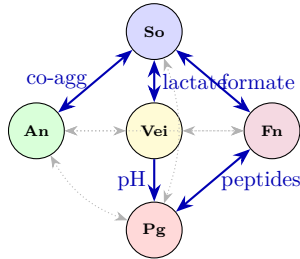
$$\boldsymbol{\theta} = (\theta_0, \dots, \theta_{14})^T \in \mathbb{R}^{15}, \quad (11)$$

organised into five biologically motivated blocks:

$$\boldsymbol{\theta} = \underbrace{(a_{11}, a_{12}, a_{22})}_{\text{So-An}} \oplus \underbrace{(a_{33}, a_{34}, a_{44})}_{\text{Vei-Fn}} \oplus \underbrace{(a_{13}, a_{14}, a_{23}, a_{24})}_{\text{cross}} \oplus \underbrace{(a_{15}, a_{25}, a_{35}, a_{45})}_{\text{Pg cross}} \oplus \underbrace{(a_{55})}_{\text{Pg}}. \quad (12)$$

2.5 Biological interaction network

Figure 1 summarises previously reported or biologically supported interactions among the five taxa, compiled from Heine et al. (2025) and the literature; it is used only to interpret the inferred parameters, not as a constraint. The listed pairs rest on different kinds of evidence (co-aggregation, metabolic cross-feeding, pH modification, peptide utilisation), which Table 1 states for each pair. Because $\mathbf{A}$ is symmetric by construction, an inferred coefficient a_{ij} describes an undirected effective interaction between species i and j ; the arrow in Figure 1 represents a metabolite flow reported in the literature, not a direction recovered by the model. All 15 entries of $\mathbf{A}$ are estimated (i.e., no $\theta_k \equiv 0$), so that no reported interaction is imposed and no unreported one is excluded.



Pair	Mechanism	Reported flow
So-An	Co-aggregation	mutual
So-Vei	Lactate exchange	So to Vei
So-Fn	Formate symbiosis	mutual
Vei-Pg	pH elevation	Vei to Pg
Fn-Pg	Peptide supply	mutual

Table 1 Previously reported or biologically supported interactions (Heine et al. 2025) used to interpret the inferred coefficients. The model coefficients a_{ij} are symmetric; the last column refers to the reported metabolite flow, not to the model. All 15 parameters are estimated; none is fixed a priori.

Fig. 1 Species interaction network (Heine et al. 2025). Solid: previously reported or biologically supported interactions considered in the interpretation of the model. Dotted grey: no reported pathway. All pairs are estimated with the same prior.

The block symbols a_{ij} follow the 1-based notation of Klempt et al. (2026); the matrix entries A_{ij} use 0-based indexing consistent with Eqs. (7)–(8).

3 Bayesian inference framework

3.1 Posterior distribution

Given observed time-course data $\mathbf{y}_{\text{obs}} = \{y_{\text{obs},i}(t_k)\}$, we seek the posterior

$$p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) = \frac{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathbf{y}_{\text{obs}})}, \quad (13)$$

where $p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})$ is the likelihood, $p(\boldsymbol{\theta})$ the prior, and $p(\mathbf{y}_{\text{obs}})$ the model evidence.

3.2 Likelihood

The observations are the species fractions at Days 3, 6, 10, 15 and 21, normalised to unit sum at each time point; Day 1 serves as the initial condition and is excluded from the likelihood. The model predicts volume fractions $\phi_i(t)$ together with a void fraction that is not observed; predictions are therefore renormalised to species fractions, $\hat{y}_i(t_k; \boldsymbol{\theta}) = \phi_i(t_k) / \sum_j \phi_j(t_k)$, before comparison. Model time is mapped linearly to experimental days. The initial condition is the normalised Day-1 observation, with a lower bound of 10^{-3} on each fraction.

For the species composition we use a weighted Gaussian log-likelihood

$$\ln p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_{k=1}^{N_t} \sum_{i=1}^5 w_{ik} \frac{(y_{\text{obs},i}(t_k) - \hat{y}_i(t_k; \boldsymbol{\theta}))^2}{\sigma_i^2} + \text{const}, \quad (14)$$

where σ_i is a species-specific observation noise derived from the replicate standard deviation of the normalised species fractions (Heine et al. 2025) ($N = 8$ biological replicates per condition), averaged over time points. The weights w_{ik} are listed in Table 2: Pg is up-weighted ($\lambda_{\text{Pg}} = 5$) because it is a low-abundance species whose late increase is the main dynamic feature of the dysbiotic data; the last two time points are up-weighted ($\lambda_{\text{late}} = 3$); and species whose mean fraction is below 5% are down-weighted ($\lambda_{\text{rare}} = 0.1$).

In Phase 2 (Section 3.3) a further Gaussian channel is added with the weight of Table 2. The *viability channel* compares the predicted community viability $\sum_i \phi_i \psi_i / \sum_i \phi_i$ with the measured fraction of membrane-intact cells (Heine et al. 2025), with σ_ψ the mean reported measurement error. The measured pH time series of the HOBIC conditions (Heine et al. 2025) is not used in the likelihood; it is kept for the check in Section 6.7. Treating the compositional data with independent Gaussian errors ignores the unit-sum constraint (Section 7).

3.3 Prior and inference stages

The prior is a product of independent uniform distributions:

$$p(\boldsymbol{\theta}) = \prod_{k=0}^{14} \frac{1}{u_k - l_k} \mathbf{1}_{[l_k, u_k]}(\theta_k), \quad (15)$$

Weight	Value	Applies to
λ_{Pg}	5	Pg, all time points
λ_{late}	3	Days 15 and 21, all species
λ_{rare}	0.1	species with mean fraction < 5%
λ_{ψ}	3.0 (DH), 2.0 (DS), 1.5 (CS, CH)	viability channel, Phase 2

Table 2 Likelihood weights.

on condition-specific boxes $[l_k, u_k]$ (Supplementary Table S1). All 15 entries of $\mathbf{A}$ are estimated in every condition; no entry is fixed a priori. The decay vector $\mathbf{b}$ does not enter the model without antibiotic treatment ($\alpha^* = 0$) and is not estimated. The initial box was set empirically to contain earlier posterior estimates; it encodes no sign constraint. Where posterior mass accumulated at a bound of this box (more than 20% of the mass in the outer 5% of the range) at any stage, the box was widened for that entry and the stage was repeated: to $[-15, 20]$ for a_{33} , a_{34} and a_{45} for Dysbiotic Static, for a_{23} , a_{24} , a_{33} , a_{34} and a_{35} for Dysbiotic HOBIC, and for a_{33} for Commensal Static, where a_{11} was also widened to $[-5, 2]$ and the upper bound of a_{13} raised to 20; the lower bounds of a_{12} , a_{14} and a_{15} were lowered to -5 for Commensal HOBIC (for this condition the posterior of a_{13} in stages (iii)–(iv) lies close to its lower bound -1 , with a 5% quantile of $[-0.93]$, below the widening threshold); the lower bounds of a_{15} , a_{23} and a_{55} were lowered to -5 for Dysbiotic Static from stage (iii) on; in addition, a_{11} and a_{14} were widened to $[-5, 5]$ and $[-3, 3]$ and the upper bound of a_{45} was raised from 6 to 20 for Dysbiotic HOBIC from stage (iii) on **[confirm with the final runs]**.

Inference proceeds in four stages that share the same data: (i) composition likelihood on the initial box; (ii) composition likelihood, box narrowed to the stage-(i) posterior mean ± 4 posterior standard deviations (SDs); (iii) composition and viability likelihood on the initial box, starting from the stage-(ii) particles (the box is not narrowed, because the likelihood changes between (ii) and (iii) and a box derived from the stage-(ii) posterior truncated the stage-(iii) posterior); (iv) as (iii), with the box narrowed to the stage-(iii) posterior mean ± 4 SDs and $N_p = 5,000$ particles. We refer to stages (i)–(ii) as Phase 1 and to stages (iii)–(iv) as Phase 2. Narrowing is clipped to the initial box; a ± 4 SD box removes less than 0.01% of a Gaussian marginal, so it shortens the tempering without truncating the posterior (a ± 2 SD box would remove about 5% of each well-determined marginal). Stages (ii)–(iv) start from a mixture of particles drawn around the previous stage’s MAP and draws from the stage prior; because the initial population is then not a sample from the prior, their log-evidence estimates are not used **[drop the $\ln \hat{Z}$ columns of Table 4 or report stage (i) only]**. Because the data are reused across stages, we report alongside the final posterior the stage-(i) posterior, which involves no data-dependent narrowing, and an inference on a much wider box ($[-15, 20]$ for all entries) with and without an additional $\mathcal{N}(0, 6^2)$ prior (Section 6.8). We monitor the fraction of posterior mass within 5% of each bound at every stage.

323 4 TMCMC inference

324 4.1 Tempered likelihood sequence

326 TMCMC (Ching and Chen 2007; Del Moral et al. 2006; Chopin 2002) constructs a
327 sequence of intermediate distributions

$$329 \quad p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta})^{\beta_m} p(\boldsymbol{\theta}), \quad 0 = \beta_0 < \beta_1 < \dots < \beta_M = 1, \quad (16)$$

331 interpolating from the prior ($\beta_0 = 0$) to the full posterior ($\beta_M = 1$). The tempering
332 parameter β_{m+1} is determined adaptively so that the coefficient of variation of the
333 importance weights

$$335 \quad w_j^{(m)} = p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}_j^{(m)})^{\beta_{m+1} - \beta_m}, \quad j = 1, \dots, N, \quad (17)$$

337 satisfies $\text{CoV}[\{w_j^{(m)}\}] = \delta_{\text{target}}$, solved by bisection on $(\beta_m, 1]$ (Betz et al. 2016).
338

340 4.2 Resampling and mutation

341 At each stage m :

- 342 1. **Resample** N particles from $\{\boldsymbol{\theta}_j^{(m)}\}$ with probabilities $\propto w_j^{(m)}$.
- 343 2. **Estimate** the weighted covariance

$$345 \quad \boldsymbol{\Sigma}^{(m)} = \sum_{j=1}^N \bar{w}_j^{(m)} (\boldsymbol{\theta}_j^{(m)} - \bar{\boldsymbol{\theta}}^{(m)}) (\boldsymbol{\theta}_j^{(m)} - \bar{\boldsymbol{\theta}}^{(m)})^T. \quad (18)$$

- 349 3. **Mutate** each particle via Metropolis–Hastings with proposal $q(\boldsymbol{\theta}^* | \boldsymbol{\theta}_j) =$
350 $\mathcal{N}(\boldsymbol{\theta}_j, \gamma^2 \boldsymbol{\Sigma}^{(m)})$ and acceptance probability

$$352 \quad \alpha = \min \left(1, \frac{p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}^*)^{\beta_{m+1}} p(\boldsymbol{\theta}^*)}{p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}_j)^{\beta_{m+1}} p(\boldsymbol{\theta}_j)} \right). \quad (19)$$

355 Proposals outside the prior box are rejected without evaluating the likelihood.
356 Random-walk proposals alternate with differential-evolution proposals (ter Braak
357 2006), $\boldsymbol{\theta}^* = \boldsymbol{\theta}_j + \gamma_{\text{DE}}(\boldsymbol{\theta}_a - \boldsymbol{\theta}_b) + \boldsymbol{\epsilon}$ with $\gamma_{\text{DE}} = 2.38/\sqrt{2d}$, which adapt to
358 strongly correlated posteriors. Each stage applies K mutation steps to every particle
359 (Table [runs]).

360 A by-product of TMCMC is the model evidence estimate

$$362 \quad \hat{p}(\mathbf{y}_{\text{obs}}) = \prod_{m=0}^{M-1} \left(\frac{1}{N} \sum_{j=1}^N w_j^{(m)} \right), \quad (20)$$

366 which can be used for Bayesian model comparison between models fitted to the same
367 data (Kass and Raftery 1995). The estimate satisfies $\ln \hat{Z} \leq \max_j \ln p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}_j)$,
368

which we check for every run. Because the likelihoods of different conditions involve different data, $\ln \hat{Z}$ is not compared across conditions.

4.3 Joint estimation in 15 dimensions

All 15 parameters are estimated jointly in every stage, made affordable by the GPU implementation of Section 5. The biologically motivated blocks (Eq. 12) serve as interpretive groups for posterior analysis but are not estimated sequentially.

4.4 Convergence assessment

Every stage is run with three independent seeds and accepted only if all of the following hold: (i) the stored log-likelihood of every particle agrees with a recomputation from the stored parameters; (ii) the tempering reaches $\beta = 1$; (iii) each particle is moved on average at least twice per stage and at least five times per free parameter over the run; (iv) the maximum log-likelihood differs by at most 1 across seeds; and (v) the posterior median of every parameter differs across seeds by at most half a posterior standard deviation. Criterion (iii) uses accepted moves rather than the acceptance rate: with optimally scaled random-walk proposals, $5d$ accepted moves displace each coordinate by about $2.38\sqrt{5} \approx 5$ posterior standard deviations, independently of d . Effective sample sizes after resampling are not reported, as they reflect the resampling rather than mixing.

5 GPU-accelerated forward model

Each TMCMC run requires $\mathcal{O}(10^5)$ – $\mathcal{O}(10^6)$ forward ODE solves (up to N_p particles $\times K$ mutation steps $\times M$ tempering stages), and every stage is repeated with three seeds. We address the resulting computational cost through GPU parallelism and a two-phase introduction of the data.

GPU-parallel TMCMC.

The forward model (Eqs. 7–10) is implemented in JAX (Bradbury et al. 2018): the implicit Euler Newton solver—including Jacobian assembly and linear solve—is compiled via `jax.jit` into a single fused XLA kernel. All N particle trajectories are evaluated simultaneously via `jax.vmap` on GPU (NVIDIA RTX 4090 for the benchmark in Table 3), replacing the $\lceil N/12 \rceil$ serial batches required on CPU. This yields a $\sim 200\times$ end-to-end speedup (Table 3).

Two-phase introduction of the data.

In Phase 1 (stages (i)–(ii), Section 3.3) the likelihood contains the species composition only. The full system (7)–(10), including the viability fractions ψ_i , is integrated in both phases; since the composition is compared after renormalisation, the viability measurements carry information only through the additional channels of Phase 2. Phase 2 (stages (iii)–(iv)) adds the viability channel and starts from the Phase 1 particles on the initial box (stage (iii)), which is then narrowed around the stage-(iii)

415 **Algorithm 1** Staged GPU-accelerated TMCMC (one seed; every stage is run with
416 three seeds and checked as in Section 4.4)

417 **Require:** Composition data $\mathbf{y}_{\text{obs}}$, viability data, N_p particles, GPU device
418 **Ensure:** MAP estimate $\hat{\boldsymbol{\theta}}^{\text{MAP}}$, posterior samples, log-evidence $\ln \hat{Z}$

- 419 1: $\triangleright$ **Stage (i): composition likelihood, initial box**
- 420 2: Draw $\{\boldsymbol{\theta}_j\}_{j=1}^{N_p}$ from the stage prior; set $\beta_0 \leftarrow 0$
- 421 3: **while** $\beta_m < 1$ **do**
- 422 4: Find β_{m+1} s.t. $\text{CoV}[\{w_j\}] = \delta_{\text{target}}$ via bisection
- 423 5: Compute weights $w_j \leftarrow p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}_j)^{\beta_{m+1} - \beta_m}$
- 424 6: Resample N_p particles $\propto w_j$ (systematic)
- 425 7: Estimate weighted covariance $\hat{\boldsymbol{\Sigma}}^{(m)}$ (Eq. 18)
- 426 8: **for** $k = 1$ **to** K **do**
- 427 9: Propose $\boldsymbol{\theta}'_j$ for all j (random-walk or differential-evolution move)
- 428 10: Evaluate all N_p likelihoods in parallel via `jax.vmap` on GPU
- 429 11: Accept/reject via MH (Eq. 19); reject proposals outside the box
- 430 12: **end for**
- 431 13: **end while**
- 432 14: Record MAP and posterior SD per parameter
- 433 15: $\triangleright$ **Stage (ii): composition likelihood, box mean ± 4 SD**
- 434 16: Repeat lines 2–13 on the narrowed box, starting from the previous particles
- 435 17: $\triangleright$ **Stages (iii)–(iv): add the viability channel; initial box, then mean**
- 436 **± 4 SD**
- 437 18: Augment likelihood with the weighted viability channel (Table 2)
- 438 19: Repeat lines 2–13 for each stage
- 439 20: **return** $\hat{\boldsymbol{\theta}}^{\text{MAP}}$, posterior samples, $\ln \hat{Z}$

442 posterior (stage (iv)). Comparing Phase 1 and Phase 2 MAP estimates shows how
443 strongly the viability data move $\mathbf{A}$.

444 The procedure is summarised in Algorithm 1; Table 3 reports the computational
445 gains.

446 A large language model (Claude, Anthropic) was used to assist with code devel-
447 opment and manuscript preparation; all code and text were reviewed and verified by
448 the authors, who take full responsibility for the content.

449

450 6 Results

451

452 The inference is performed under four experimental conditions from Heine et al.
453 (2025): Commensal Static (CS), Commensal HOBIC (CH), Dysbiotic Static (DS),
454 and Dysbiotic HOBIC (DH). Species distribution data ($N = 8$ biological replicates
455 per condition, Days 1, 3, 6, 10, 15, 21) were normalised to unit sum at each time
456 point; Day 1 serves as initial condition. All 15 interaction parameters are estimated
457 simultaneously for every condition, following the staged procedure of Section 3.3 and
458 Algorithm 1. Each stage is run with three seeds; Table [runs] lists particles, muta-
459 tion steps and number of tempering stages per condition, together with the outcome
460

Metric	CPU (12 cores)	GPU (RTX 4090)	Speedup
<i>Matched benchmark (DH, $N_p = 500$, $K = 10$, 9 stages):</i>			
Per stage	2 400 s	14 s	170×
Total	21 600 s	147 s	147×
<i>Final stage (iv) (4 cond., $N_p = [n]$, $K = [n]$, GPU <i>[model]</i>):</i>			
Per condition	[h] [†]	[h]	[x]
All 4 cond.	[h] [†]	[h]	[x]

Table 3 Computational cost comparison. The matched benchmark (500 particles, identical settings) isolates the GPU speedup at about 150–170× (170× per stage, 147× in total). The final-stage times are measured per seed (**update with the final runs**). [†]Extrapolated from the 500-particle CPU benchmark assuming linear scaling in N_p .

of the convergence criteria of Section 4.4. All runs use a coefficient-of-variation target $\delta_{\text{target}} = 1.0$ for the adaptive β -schedule (Betz et al. 2016) and an initial random-walk scale $\gamma^2 = 2.38^2/d$ (Roberts and Rosenthal 2001). The random-walk scale is adapted online after each random-walk step: γ^2 is multiplied by 0.5 when the acceptance rate of that step is below 10%, by 0.8 when it is below 18%, and by 1.3 when it exceeds 35%; the differential-evolution scale is kept fixed.

[All numbers in this section are from the previous runs and will be replaced by the final stage-(iv) runs.]

6.1 Quantitative fit metrics

Table 4 reports the composition fit at the end of Phase 1 (composition likelihood only) and Phase 2 (with the viability channel). Phase 1 fits the composition at least as well as Phase 2 in every condition, as expected since Phase 2 also has to match the viability data. In Phase 2, DS has the best composition fit (RMSE=0.033, $R_{Pg}^2 = 0.85$), and DH reproduces the late Pg increase ($R_{Pg}^2 = 0.90$). The composition RMSE increases by at most 0.035 from Phase 1 to Phase 2.

Cond.	Phase 1: composition only					Phase 2: with viability — final				
	RMSE	MAE	$\bar{\alpha}$	$\max \ln \mathcal{L}$	$\ln \hat{Z}$	RMSE	MAE	$\bar{\alpha}$	$\max \ln \mathcal{L}$	$\ln \hat{Z}$
CS	0.119	0.077	0.14	−6.3	−2.6	0.119	0.075	0.12	−21.6	−3.3
CH	0.071	0.053	0.17	−3.6	−1.3	0.104	0.088	0.07	−112.3	−8.5
DS	0.020	0.025	0.11	−1.2	−3.8	0.033	0.024	0.13	−93.4	−2.8
DH	0.067	0.045	0.10	−6.6	−9.6	0.087	0.060	0.07	−170.4	−11.9

Table 4 Composition fit at the end of Phase 1 (stage (ii), composition likelihood only) and Phase 2 (stage (iv), with the viability channel). RMSE and MAE refer to the species composition in both phases. $\bar{\alpha}$: mean MH acceptance rate. $\max \ln \mathcal{L}$ and $\ln \hat{Z}$ are not comparable across phases (Phase 2 includes additional channels) nor across conditions (different data); use RMSE for comparison.

Table 5 reports the per-species R^2 for each condition. In commensal states (CS, CH), Fn and Pg fractions are near zero throughout, rendering R^2 undefined (total variance ≈ 0); these are marked “—” in the table. In dysbiotic states, in Phase 1, DH achieves $R^2 > 0.8$ for An , Vei , and Pg ; in Phase 2, Vei drops to 0.63 while An (0.86) and Pg (0.90) remain above 0.8. DS achieves $R^2 = 0.93$ for Vei in Phase 1 (0.91 in Phase 2) and 0.85 for Pg in both phases. So shows negative R^2 in most conditions in both phases; in Phase 2 this partly reflects a trade-off between the composition and the viability channel. So also exhibits the highest replicate variability ($\sigma_{So} \approx 0.19$), which inflates SS_{tot} and further depresses R^2 .

	Phase 1 (composition only)					Phase 2 (with viability) — final				
Cond.	So	An	Vei	Fn	Pg	So	An	Vei	Fn	Pg
CS	-0.39	0.80	-3.2	—	—	-0.36	0.91	-3.7	—	—
CH	0.55	0.70	0.59	—	—	0.40	-0.22	-0.14	—	—
DS	-0.87	0.76	0.93	0.54	0.85	-0.86	0.73	0.91	0.66	0.85
DH	-0.47	0.83	0.83	0.60	0.92	-0.21	0.86	0.63	0.06	0.90

Table 5 Per-species R^2 of the composition for Phase 1 and Phase 2. “—”: species fraction < 0.02 at all time points ($SS_{tot} \approx 0$), rendering R^2 ill-defined; the MAP prediction for these species is < 0.01 with absolute error below the observation noise σ_i . Bold: $R^2 > 0.8$. DS maintains $R^2 \geq 0.85$ for Vei and Pg in both phases; DH reproduces the Pg time course ($R^2 = 0.90$) in Phase 2.

6.2 Convergence diagnostics

All stages of all conditions satisfy the convergence criteria of Section 4.4 [table of runs: seeds, moves per parameter, spread of max log-likelihood and medians]. The acceptance rates of $[x-y]\%$ are below the optimum $\bar{\alpha} \approx 23\%$ for Gaussian targets (Roberts and Rosenthal 2001), reflecting the non-Gaussianity and bounded support of the posterior; criterion (iii) therefore counts accepted moves rather than relying on the acceptance rate.

6.3 Posterior fits across conditions

Figure 2 presents the posterior predictive fits for all four conditions in the order of increasing dysbiotic severity (columns left to right); each row corresponds to one species. The solid line shows the MAP trajectory of φ_i (total volume fraction); the dashed line shows $\bar{\varphi}_i = \varphi_i \psi_i$ (living fraction), which differs from φ_i only where the viability $\psi_i < 1$. In CS (column 1), the model reproduces the initial rapid growth and steady state of So and An ; pathogen species remain near zero—not due to locking, but because the posterior concentrates these entries near zero. In CH (column 2), the So -dominated “blue bloom” under flow is correctly captured. DS (column 3) achieves the best fit (RMSE = 0.033) with clear pathogen succession under static cultivation. In DH (column 4), the model captures the non-linear amplification of Vei and Pg , including the late increase of Pg . The credible bands widen from CS to DH, reflecting the

increasing effective dimensionality. The bottom row highlights the distinct *Pg* strains: the avirulent DSM20709 used in commensal conditions remains near zero, whereas the virulent W83 in dysbiotic conditions increases late. Similarly, the *Veillonella* row corresponds to *V. dispar* in commensal and *V. parvula* in dysbiotic conditions (Heine et al. 2025).

6.4 Inferred interaction matrices

The MAP interaction matrices (Figure 3) reveal the progressive reorganisation of the interaction landscape from commensal to dysbiotic states. In CS and CH, the dominant entries are the *So–An* and *So–Vei* blocks, reflecting the mutualistic core of early colonizers; the *Fn* and *Pg* rows are near zero. In DS, pathogen-related entries appear at moderate magnitude. In DH, the effective *Vei–Pg* and *Fn–Pg* coefficients are positive. The MAP values of the Pg-related parameters are $\hat{a}_{55}^{\text{DH}} = [3.43]$ (Pg self), $\hat{a}_{45}^{\text{DH}} = [5.63]$ (Fn–Pg) and $\hat{a}_{44}^{\text{DH}} = [4.45]$ (Fn self). In DS, these coefficients are [weaker] ($\hat{a}_{55} = [0.30]$, $\hat{a}_{45} = [2.20]$, $\hat{a}_{44} = [0.39]$), while in CS and CH they are not distinguishable from zero.

6.5 Posterior parameter distributions

Figure 4 shows the posterior distributions of the 15 interaction matrix entries across all four conditions. In commensal conditions (CS, CH), the Pg-related parameters (a_{15} , a_{25} , a_{35} , a_{45} , a_{55}) are not distinguishable from zero, although none of them is fixed a priori. In dysbiotic conditions (DS, DH), several of these parameters shift to positive values with broader distributions [check which with final runs]. DS exhibits the narrowest posteriors overall [check with final runs].

6.6 Quantitative comparison across conditions

To quantify the inter-condition differences, we compute for each pair (c_1, c_2) the correlation coefficient and relative Frobenius distance:

$$\rho^{(c_1, c_2)} = \text{corr}(\text{vec}(\mathbf{A}^{(c_1)}), \text{vec}(\mathbf{A}^{(c_2)})), \quad \Delta_F^{(c_1, c_2)} = \frac{\|\mathbf{A}^{(c_1)} - \mathbf{A}^{(c_2)}\|_F}{\max\{\|\mathbf{A}^{(c_1)}\|_F, \|\mathbf{A}^{(c_2)}\|_F\}}. \quad (21)$$

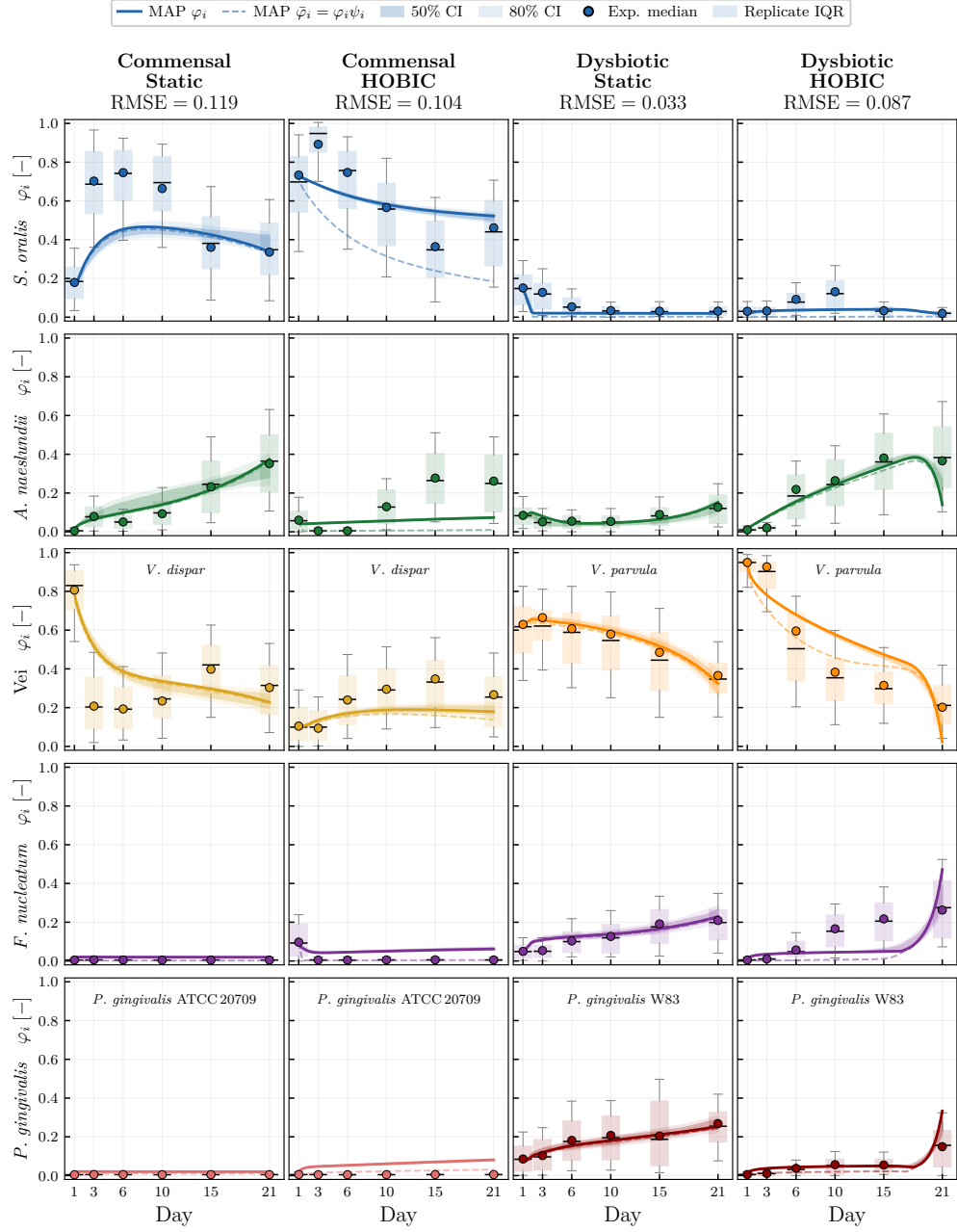


Fig. 2 Posterior predictive fits of volume fraction ϕ_i (Phase 2, stage (iv)). Columns: four experimental conditions (CS, CH, DS, DH); rows: five species. Solid lines: MAP ϕ_i trajectory; dashed lines: living fraction $\bar{\phi}_i = \phi_i \psi_i$. Shaded bands: 50% (dark) and 80% (light) credible intervals from 100 posterior samples. Circles: experimental median ($N = 8$ replicates); box plots: inter-replicate variability (IQR). Note that *P. gingivalis* employs different strains: avirulent DSM 20709 in commensal conditions and virulent W83 in dysbiotic conditions (indicated by colour). *Veillonella* species also differ: *V. dispar* (commensal) and *V. parvula* (dysbiotic) (Heine et al. 2025).

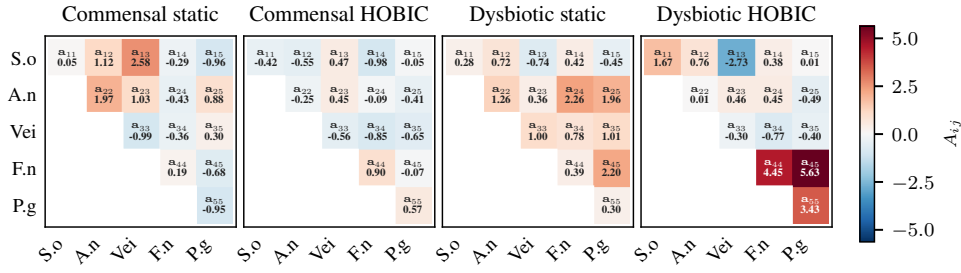


Fig. 3 MAP interaction matrices $\mathbf{A}$ (Phase 2, stage (iv)). Red: cooperative ($A_{ij} > 0$); blue: competitive ($A_{ij} < 0$). In commensal conditions (CS, CH), the Fn and Pg rows/columns are near zero; no entry is fixed a priori. The Fn–Pg coefficient is positive in the dysbiotic conditions.

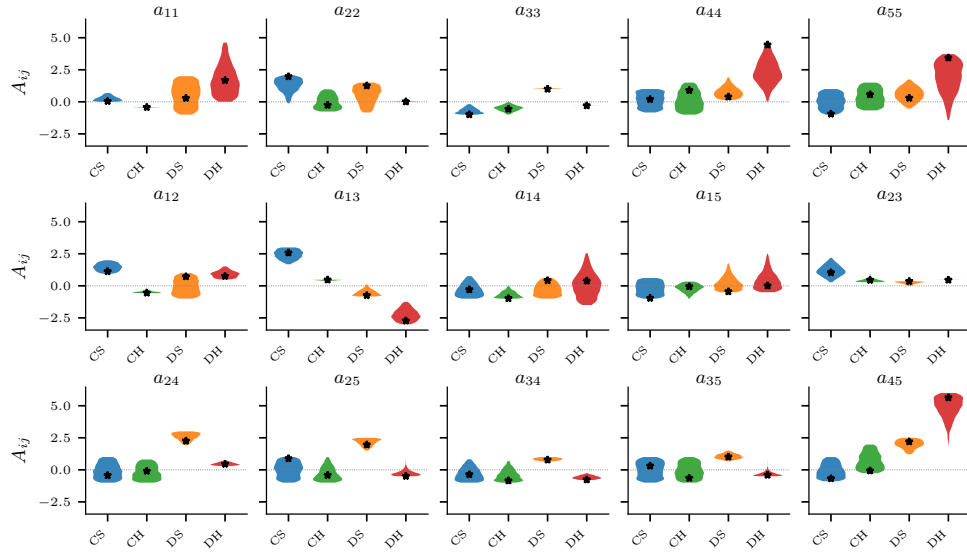


Fig. 4 Posterior distributions of the 15 interaction matrix entries (Phase 2, stage (iv)). Stars: MAP estimates. Whether a narrow marginal reflects the data or the prior box is assessed in Section 6.8.

C1	C2	ρ	Δ_F	d_{15D}
CH	CS	+0.26	1.06	3.38
DH	DS	+0.40	0.88	5.86
CH	DH	+0.24	1.01	6.33
CH	DS	-0.27	1.27	4.88
DH	CS	-0.49	1.28	7.77
CS	DS	-0.27	1.36	5.88

Table 6 Pairwise comparison of MAP interaction matrices. ρ : correlation of vectorised $\mathbf{A}$; Δ_F : relative Frobenius distance; d_{15D} : posterior centroid distance in 15D A-matrix space.

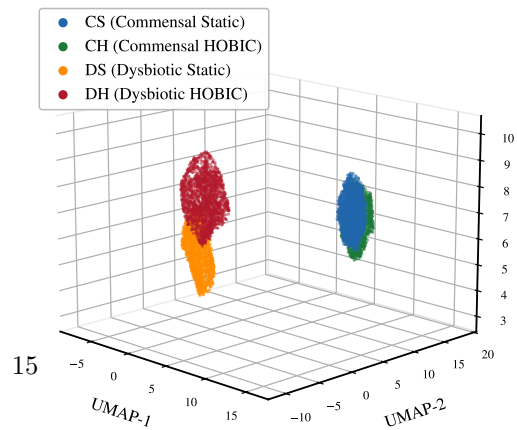


Fig. 5 UMAP 3D embedding of posterior A-matrix samples (Phase 2, stage (iv)). The four conditions form well-separated clusters.

The results (Table 6) show that matrices within the same health state share a common backbone ($\rho_{\text{CH-CS}} = +0.26$, $\rho_{\text{DH-DS}} = +0.40$), whereas cross-state comparisons yield negative correlations ($\rho_{\text{DH-CS}} = -0.49$) and large Frobenius distances ($\Delta_F \geq 0.88$). The UMAP embedding of the full posterior (Figure 5) confirms that all four conditions form well-separated clusters with no overlap (McInnes et al. 2018); centroid distances in 15D space mirror the MAP ordering (CS–CH closest at $d=3.38$; CS–DH most distant at $d=7.77$). Overall, the transition from health to disease corresponds to a structural reorganisation of the interaction landscape, not merely a rescaling of existing coefficients.

6.7 Post hoc consistency checks

We compare the inferred trajectories with measurements of Heine et al. (2025) that are related to, but not identical with, the composition data. These comparisons are consistency checks rather than independent validation. pH ($R^2 = [0.78]$, RMSE = $[0.13]$ pH units): the pH time series of the HOBIC conditions is not used in the inference. A linear relation between pH and the measured species fractions is fitted once, by ordinary least squares on the experimental data alone ($N = 90$ HOBIC time points, commensal and dysbiotic combined; leave-one-out RMSE = 0.056), selecting the predictors by leave-one-out cross-validation:

$$\text{pH}(t) = 6.10 + 0.16 \phi_{\text{So}} + 0.55 \phi_{\text{An}} + 0.30 \phi_{\text{Vei}} + 1.22 \phi_{\text{Fn}} + 1.88 \phi_{\text{Pg}}. \quad (22)$$

Applying Eq. (22) with fixed coefficients to the species fractions predicted from [500] posterior samples of stage (iv) reproduces the measured pH of CH and DH with $R^2 = [0.78]$ ([12] points). The pH data enter only the fit of Eq. (22) on measured fractions, never the inference; the check therefore tests whether the inferred model reproduces the species fractions that drive the pH, not the pH mechanism itself.

Gingipain–Pg correlation ($r = [0.90]$): the temporal profile of gingipain (Heine et al. 2025, Fig. 4B), the principal virulence factor of *Pg*, correlates with the predicted Pg trajectory in DH, both increasing at Day 15–21; this is a temporal correlation, not a test of a mechanism.

Metabolic sign consistency: the MAP *So–An* coefficient ($a_{12} = [+0.76]$) is consistent with co-aggregation and cross-feeding between these early colonisers, while the *So–Vei* coefficient is negative ($a_{13} = [-2.73]$); this is a property of the effective model and should not be read as evidence against the well-established lactate cross-feeding between streptococci and *Veillonella*.

6.8 Transferability and identifiability

Cross-condition transferability.

To test whether the inferred matrices capture genuine condition-specific structure, we evaluate each MAP on every other condition’s data (Table 7). Within-state commensal transfer (CS to CH) achieves an RMSE of 0.098, *below* the CH self-prediction of 0.104, indicating a shared commensal interaction backbone that generalises across

cultivation modes. Cross-state transfers (DS to CS: 0.384, CS to DH: 0.198) degrade sharply, indicating that commensal and dysbiotic matrices differ.

Source \ Target	CS	CH	DS	DH
CS	0.119	0.098	0.261	0.198
CH	0.257	0.104	0.100	0.159
DS	0.384	0.259	0.033	0.255
DH	0.342	0.212	0.289	0.087

Table 7 Cross-condition prediction RMSE. Rows: condition whose MAP is used; columns: data. Diagonal (bold): self-prediction, equal to the calibrated RMSE.

Identifiability.

The ratio $r_i = \sigma_{\text{post},i} / \Delta_{\text{prior}}$ cannot detect non-identifiability: a posterior identical to a uniform prior already gives $r = 1/\sqrt{12} \approx 0.29$. We therefore assess identifiability by (i) agreement across independent seeds (Section 4.4), (ii) the dependence of each marginal on the prior, comparing inference on the wide box $[-15, 20]$ with and without an $\mathcal{N}(0, 6^2)$ prior, and (iii) the fraction of posterior mass at the box bounds. **[Result: which entries are determined by the data and which depend on the prior (DH and DS).]** Parameters whose marginals change with the prior are reported as weakly identified, and no biological interpretation is based on them.

Likelihood weights.

The weights $\lambda_{\text{Pg}} = 5$ and $\lambda_{\text{late}} = 3$ (Table 2) are modelling choices. Repeating stage (i) for DH with $\lambda_{\text{Pg}} = \lambda_{\text{late}} = 1$ (three seeds, all convergence criteria met) leaves the median of a_{45} unchanged (+3.0 in all seeds, compared with +2.9 to +3.2 with the weights), but widens its 90% interval so that the lower end reaches zero ($[-0.06, +5.6]$ to $[+0.17, +5.6]$, compared with $[+0.5, +5.5]$ to $[+1.1, +5.3]$). The predicted Pg increase from Day 15 to Day 21 becomes less consistent across seeds (ratio 1.7–3.1, compared with 2.4–2.9 with the weights; measured 2.74), and the composition RMSE changes little (0.059–0.068 compared with 0.058–0.065). The weights thus sharpen the estimate of the Fn–Pg coefficient and stabilise the late Pg increase, but do not create the positive coefficient.

Necessity of the Fn–Pg interaction.

We compare the full model with a reduced model in which $a_{45} = 0$, both inferred from stage (i), whose particles are initialised from the prior so that the log-evidence estimate is valid. **[Table: $\ln \hat{Z}$ (mean $\pm$ spread over three seeds) of both models and $\ln B = \ln \hat{Z}_{\text{full}} - \ln \hat{Z}_{a_{45}=0}$ for the four conditions.]** Because the prior on a_{45} is uniform, $\ln B$ includes an Occam penalty proportional to the log of the box width and is therefore conservative for the full model. The box widths differ between conditions (Supplementary Table S1): for DS, where the box had to be widened to $[-15, 20]$

(Section 3.3), the penalty is $\ln 35 \approx 3.6$, compared with $\ln 6.5 \approx 1.9$ for DH. We therefore read $\ln B$ only within each condition, and compare conditions through the posterior of a_{45} rather than through $\ln B$. In DS, a_{45} and a_{33} trade off: with $a_{45} = 0$ the posterior moves almost entirely to the mode $a_{33} \approx -10$ with nearly the same fit (maximum log-likelihood **[-1.5 versus -0.6]**), whereas the full model places most of its mass at $a_{33} \approx +1$ with $a_{45} > 0$. The relative weight of the two a_{33} modes is not reliably determined by the data, and we therefore report a_{45} for DS separately for each mode: in the mode $a_{33} \approx +1$, which carries most of the mass and fits better (maximum log-likelihood higher by 0.6–1.2), $a_{45} = \mathbf{[+2.4]}$ with a 90% interval excluding zero (**[[+0.5, +4.4]]**); in the mode $a_{33} \approx -10$, $a_{45} = \mathbf{[+0.8]}$ with an interval that includes zero (**[[−0.2, +1.4]]**). The evidence for $a_{45} > 0$ in DS is therefore conditional on the a_{33} mode, unlike in DH.

7 Discussion

Biological interpretation.

The inferred interaction matrices provide a quantitative, effective description of the commensal and dysbiotic community states. Under commensal conditions, the dominant coefficients involve the early colonisers *So*, *An* and *Vei*, consistent with their reported co-aggregation and cross-feeding. Under dysbiotic conditions, the effective *Vei*–Pg and *Fn*–Pg coefficients become positive, with magnitudes that depend on the cultivation regime, whereas the Pg-related coefficients in both commensal conditions are not distinguishable from zero. Differences between commensal and dysbiotic matrices combine the effect of community state with those of the *Veillonella* species (*V. dispar* vs. *V. parvula*) and the Pg strain (DSM 20709 vs. W83), and cannot be attributed to the community state alone.

Matrices within the same community state are positively correlated, whereas cross-state comparisons are **[anticorrelated]** (Table 6). Together with the cross-condition transfer (Table 7), in which MAP estimates transfer within community states across cultivation modes but not across states, this indicates that the commensal and dysbiotic data require different interaction matrices rather than a rescaling of a shared one **[re-check both tables with the final runs]**.

A testable prediction.

Beyond reproducing the observed community states, the inferred model yields a specific prediction. In posterior predictive simulations of the dysbiotic dynamic condition, the late Pg increase (Day-21/Day-15 ratio ≥ 1.5) occurs in **[99–100]**% of posterior samples. Removing *F. nucleatum* from the consortium (its volume fraction held at zero) reduces this to **[10–13]**%, and setting only the Fn–Pg coefficient a_{45} to zero reduces it to **[2–6]**%. The late increase thus arises in the model from a delayed rise of *F. nucleatum* transmitted to Pg through a_{45} , consistent with the bridging role of *F. nucleatum* (Kolenbrander et al. 2010). The model therefore predicts, with posterior probability **[0.87–0.90]**, that a leave-*F. nucleatum*-out co-culture under otherwise identical dynamic cultivation suppresses the late Pg increase. Because commensal and dysbiotic consortia differ in *Veillonella* species and Pg strain as well as in community

state, the prediction concerns this consortium and should not be generalised to other strain combinations.

Effect of the viability data.

Figure 6 compares the Phase 1 (composition only) and Phase 2 (with viability) MAP estimates parameter by parameter. In commensal conditions, the two phases yield consistent MAPs ($r = [0.83]–[0.90]$): the additional data change **A** little. In dysbiotic conditions, the MAPs differ more ($r = [0.23]$ for DS, $[0.62]$ for DH), with the largest shifts in Pg-related parameters, where the measured viability changes most. The viability predictions themselves remain approximate (RMSE $[0.11–0.39]$), indicating that the ψ -dynamics of the model capture the loss of membrane integrity only coarsely; the viability channel acts mainly as an additional constraint on **A**.

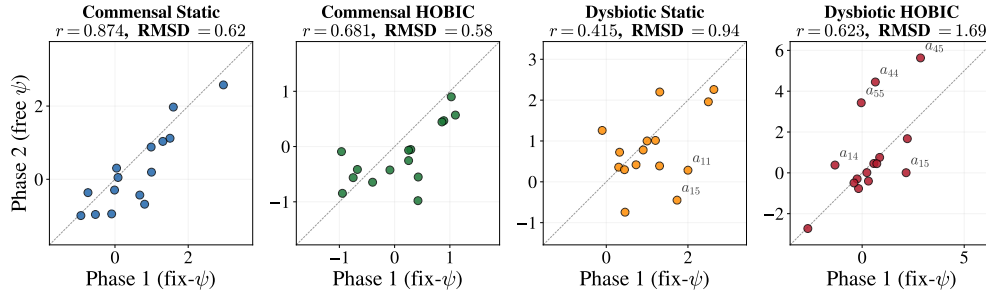


Fig. 6 Phase 1 (composition only) vs. Phase 2 (with viability) MAP estimates. Each point is one of the 15 parameters; dashed line: identity. **[update with the final runs]**

Limitations.

The likelihood treats the five species fractions as independent Gaussian observations although they sum to one. The constraint couples the residuals, so the effective number of independent observations is smaller than the number of data points and the posterior intervals may be somewhat too narrow. A likelihood on log-ratios would respect the constraint, but several fractions in the data are close to zero, where log-ratios are unstable; we therefore keep the Gaussian form, which is common in calibrations of continuum biofilm models. The interaction coefficients are effective quantities of this model and data set; they summarise the net effect of mechanisms that the model does not resolve, such as metabolite exchange and pH, and should not be read as direct measurements of pairwise mechanisms.

Computational considerations.

The $\sim 150\times$ GPU speedup (Table 3) exploits the embarrassingly parallel mutation step via `jax.vmap`. For this solver and workload, gradient-based samplers (HMC, NUTS) were less efficient on the GPU, because trajectories of variable length limit batched

875 evaluation (Dance et al. 2025). Combining the direct-ODE approach with surrogate
876 acceleration (Fritsch et al. 2026) is a natural extension.

877

878 8 Conclusions

879

880 We have presented a GPU-accelerated Bayesian framework for estimating the
881 interaction parameters of a five-species oral biofilm model. The key contributions are:

- 882 1. **Staged inference with explicit convergence checks.** The interaction matrix
883 is first inferred from the species composition and then refined with the viability
884 data; every stage is run with three seeds and accepted only if explicit convergence
885 criteria hold. All 15 parameters are estimated without fixing any of them a priori;
886 the Pg-related coefficients in the commensal conditions are not distinguishable from
887 zero.
- 888 2. **GPU-accelerated TMCMC.** Compiling the forward model via JAX and batch-
889 ing all particles on GPU achieves a $\sim 150\times$ speedup (Table 3), making repeated
890 runs with up to $N_p = 5,000$ particles feasible.
- 891 3. **Reproduction of four conditions and a testable prediction.** The frame-
892 work reproduces commensal homeostasis (CS, CH), moderate dysbiosis (DS), and
893 the late Pg increase (DH). Posterior predictive simulations predict that removing
894 *F. nucleatum* suppresses the late Pg increase under dysbiotic dynamic cultivation,
895 which can be tested by a leave-one-species-out co-culture.
- 896 4. **Quantitative interaction comparison.** Pairwise comparison and cross-
897 condition transfer indicate that commensal and dysbiotic data require different
898 interaction matrices, whereas matrices within a community state transfer across
899 cultivation modes.
- 900 5. **Identifiability.** Identifiability is assessed by seed agreement, prior dependence on
901 a wide box and the posterior mass at the bounds, rather than by the posterior-to-
902 prior width ratio, which cannot detect non-identifiability; **[summary of which**
903 **entries are identified]**.

904

905 *Future work.*

- 906 • **Experimental test of the prediction:** a leave-*F. nucleatum*-out co-culture under
907 dysbiotic dynamic cultivation.
- 908 • **External validation:** testing the inferred interactions against independent in-vivo
909 datasets (e.g., Dieckow et al. (2024)).
- 910 • **Spatially resolved inference:** coupling the framework to the continuum PDE
911 biofilm model (Klempt et al. 2026) to exploit depth-resolved FISH/CLSM imaging.

912

913 **Acknowledgements.** The authors acknowledge the Lower Saxony Centre for
914 Biomedical Engineering, Implant Research and Development (NIFE), Hannover. Com-
915 putational resources of the Multiphysics Materials Computation Research Group
916 (Assoc. Prof. M. Muramatsu), Department of Mechanical Engineering, Keio Univer-
917 sity, are gratefully acknowledged.

918

919

920

Statements and Declarations

Funding. This work was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under the Collaborative Research Center SFB/TRR-298-SIIRI – Project ID 426335750. Funded by the European Union (ERC, Gen-TSM, project number 101124463). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

Competing interests. The authors declare no competing interests.

Ethics approval and consent to participate. Not applicable. This study analyses previously published in-vitro data and involves no human participants or animals.

Data availability. This study analyses previously published data. The experimental time courses were reported by Heine et al. (2025) and are available from that publication; they are not redistributed with this article.

Code availability. Code, run configurations, convergence records and posterior samples are archived at Zenodo ([DOI]) and developed at <https://github.com/keisuke58/Tmcmc202601>.

Author contributions. [draft, to be confirmed by all co-authors] K.N. and P.J. conceived the study. K.N. developed the inference framework, implemented the GPU/JAX forward model and the TMCMC sampler, performed the analyses, and drafted the manuscript. F.K. and H.G. contributed the continuum biofilm model and its numerical formulation; M.So. contributed to the variational formulation and numerical methods. N.H. designed and performed the in-vitro biofilm experiments and curated the time-course data. K.D.-N. provided imaging resources and data and contributed to data interpretation. R.M. and S.P.S. contributed experimental data and biological interpretation. M.St. provided clinical context and resources. P.J. supervised the study and acquired funding. All authors critically reviewed and approved the final manuscript.

References

- Bradbury, J., Frostig, R., Hawkins, P., Johnson, M.J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., Zhang, Q.: JAX: Composable transformations of Python+NumPy programs (2018). <https://github.com/google/jax>
- Betz, W., Papaioannou, I., Straub, D.: Transitional Markov chain Monte Carlo: Observations and improvements. *Journal of Engineering Mechanics* **142**(5), 04016016 (2016)

- Ching, J., Chen, Y.-C.: Transitional Markov chain Monte Carlo method for Bayesian model updating, model class selection, and model averaging. *Journal of Engineering Mechanics* **133**(7), 816–832 (2007)
- Chopin, N.: A sequential particle filter method for static models. *Biometrika* **89**(3), 539–551 (2002) <https://doi.org/10.1093/biomet/89.3.539>
- Cui, Z., Wang, P., Gao, W.: Microbial dysbiosis in periodontitis and peri-implantitis: Pathogenesis, immune responses, and therapeutic. *Frontiers in Cellular and Infection Microbiology* **15**, 1517154 (2025) <https://doi.org/10.3389/fcimb.2025.1517154>
- Dance, H., Glaser, P., Orbanz, P., Adams, R.P.: Efficiently vectorized MCMC on modern accelerators. In: *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. *Proceedings of Machine Learning Research*, vol. 267, pp. 12436–12458. PMLR, ??? (2025). <https://proceedings.mlr.press/v267/dance25a.html>
- Del Moral, P., Doucet, A., Jasra, A.: Sequential Monte Carlo samplers. *Journal of the Royal Statistical Society: Series B* **68**(3), 411–436 (2006)
- Dieckow, S., Szafranski, S.P., Grischke, J., Qu, T., Doll-Nikutta, K., Steglich, M., Yang, I., Häussler, S., Stiesch, M.: Structure and composition of early biofilms formed on dental implants are complex, diverse, subject-specific and dynamic. *npj Biofilms and Microbiomes* **10**(1), 155 (2024) <https://doi.org/10.1038/s41522-024-00624-3>
- Fritsch, L., Geisler, H., Grashorn, J., Klempt, F., Soleimani, M., Broggi, M., Junker, P., Beer, M.: Bayesian Updating of constitutive parameters under hybrid uncertainties with a novel surrogate model applied to biofilms. *Computational Mechanics* **78**(2), 761–782 (2026) <https://doi.org/10.1007/s00466-026-02777-8>
- Hackl, K.: Generalized standard media and variational principles in classical and finite elastoplasticity. *Journal of the Mechanics and Physics of Solids* **45**(5), 667–688 (1997) [https://doi.org/10.1016/S0022-5096\(96\)00110-X](https://doi.org/10.1016/S0022-5096(96)00110-X)
- Heine, N., Bittroff, K., Szafranski, S.P., Duitscher, M., Behrens, W., Vollmer, C., Mikolai, C., Kommerein, N., Debener, N., Frings, K., Heisterkamp, A., Scheper, T., Torres-Mapa, M.L., Bahnemann, J., Stiesch, M., Doll-Nikutta, K.: Influence of species composition and cultivation condition on peri-implant biofilm dysbiosis in vitro. *Frontiers in Oral Health* **6**, 1649419 (2025) <https://doi.org/10.3389/froh.2025.1649419>
- Junker, P., Balzani, D.: An extended Hamilton principle as unifying theory for coupled problems and dissipative microstructure evolution. *Continuum Mechanics and Thermodynamics* **33**(4), 1931–1956 (2021) <https://doi.org/10.1007/s00161-021-01017-z>
- Junker, P., Bode, T., Hackl, K.: On a holistic variational formulation for material modeling including dissipative evolution. *Journal of the Mechanics and Physics of*

Solids 200 , 106133 (2025) https://doi.org/10.1016/j.jmps.2025.106133	1013
Junker, P., Makowski, J., Hackl, K.: The principle of the minimum of the dissipation potential for non-isothermal processes. <i>Continuum Mechanics and Thermodynamics</i> 26 (3), 259–268 (2014) https://doi.org/10.1007/s00161-013-0299-4	1014
Klempt, F., Geisler, H., Soleimani, M., Junker, P.: A continuum multi-species bacterial growth model with a novel interaction scheme. <i>Archive of Applied Mechanics</i> 96 (8), 164 (2026) https://doi.org/10.1007/s00419-026-03160-y	1015
Kolenbrander, P.E., Palmer Jr., R.J., Periasamy, S., Jakubovics, N.S.: Oral multi-species biofilm development and the key role of cell–cell distance. <i>Nature Reviews Microbiology</i> 8 (7), 471–480 (2010)	1016
Kass, R.E., Raftery, A.E.: Bayes factors. <i>Journal of the American Statistical Association</i> 90 (430), 773–795 (1995) https://doi.org/10.1080/01621459.1995.10476572	1017
Klempt, F., Soleimani, M., Wriggers, P., Junker, P.: A Hamilton principle-based model for diffusion-driven biofilm growth. <i>Biomechanics and Modeling in Mechanobiology</i> 23 (6), 2091–2113 (2024) https://doi.org/10.1007/s10237-024-01883-x	1018
Marsh, P.D.: Are dental diseases examples of ecological catastrophes? <i>Microbiology</i> 149 (2), 279–294 (2003) https://doi.org/10.1099/mic.0.26082-0	1019
McInnes, L., Healy, J., Melville, J.: UMAP: Uniform manifold approximation and projection for dimension reduction. <i>arXiv preprint arXiv:1802.03426</i> (2018)	1020
Robert, C.P., Casella, G.: <i>Monte Carlo Statistical Methods</i> , 2nd edn. Springer, ??? (2004)	1021
Roberts, G.O., Rosenthal, J.S.: Optimal scaling for various Metropolis–Hastings algorithms. <i>Statistical Science</i> 16 (4), 351–367 (2001) https://doi.org/10.1214/ss/1015346320	1022
Braak, C.J.F.: A Markov chain Monte Carlo version of the genetic algorithm Differential Evolution: easy Bayesian computing for real parameter spaces. <i>Statistics and Computing</i> 16 (3), 239–249 (2006) https://doi.org/10.1007/s11222-006-8769-1	1023
	1024
	1025
	1026
	1027
	1028
	1029
	1030
	1031
	1032
	1033
	1034
	1035
	1036
	1037
	1038
	1039
	1040
	1041
	1042
	1043
	1044
	1045
	1046
	1047
	1048
	1049
	1050
	1051
	1052
	1053
	1054
	1055
	1056
	1057
	1058